

Algorithm for file updates in Python

Project description

Access to restricted content is controlled by an allowed list of IP addresses. I created a Python algorithm that checks the allow list and removes IP addresses that should no longer have access.

Open the file that contains the allow list

```
import_file = "allow_list.txt"  
with open(import_file, "r") as file:
```

I stored the file name in `import_file`. The `with` statement opens the file, and `"r"` means read mode. `with` also closes the file automatically when done.

Read the file contents

```
ip_addresses = file.read()
```

`.read()` reads the whole file as one string and stores it in `ip_addresses`.

Convert the string into a list

```
ip_addresses = ip_addresses.split()
```

`.split()` breaks the string into a list of individual IP addresses, so I can work with each one separately.

Iterate through the remove list

```
for element in ip_addresses:
```

The `for` loop goes through each IP address in the list one at a time. `element` holds the current IP address.

Remove IP addresses that are on the remove list

```
if element in remove_list:  
    ip_addresses.remove(element)
```

The if statement checks whether the current IP address is in `remove_list`. If it is, `.remove()` deletes it from the allow list.

Update the file with the revised list of IP addresses

```
ip_addresses = "".join(ip_addresses)  
with open(import_file, "w") as file:  
    file.write(ip_addresses)
```

`.join()` turns the list back into a string. Opening the file with "w" replaces its contents, and `.write()` saves the updated allow list.

Summary

I created an algorithm that opens the allow list, converts it into a list of IP addresses, removes any address in the remove list, and saves the updated file. I also put these steps into a function, `update_file()`, so it can be reused.